

FIRECODE: A Computational Chemistry Workflow Driver and Modular Hub

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

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Submitted: 01 January 1970

Published: unpublished

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Summary

Computational chemistry, which seeks to model molecular systems for the extraction of properties of interest, is experiencing a significant renaissance thanks to impressive advances in theoretical models enabling ever-faster simulations of complex processes via density functional theory (DFT), semiempirical, and machine-learned interatomic potentials (MLIPs). However, the reduced computational cost of these workflows will only translate into faster and more robust modeling with the help of automation frameworks, where trivial human intervention does not become the new bottleneck of computational pipelines. FIRECODE aims at filling this need by providing a modular framework for generating and processing molecular ensembles.

Statement of need

FIRECODE is a modular workflow driver for computational chemistry written in Python. The code interfaces with a series of fast modern calculators (xTB ([Bannwarth et al., 2019](#)), tblite ([GitHub - Tblite/Tblite; Katbashev et al., 2025](#)), AIMNET2 ([Anstine et al., 2024](#)), UMA ([Wood et al., 2025](#)), usually via ASE ([Ase / Ase · GitLab — Gitlab.com; Bahn & Jacobsen, 2002; Larsen et al., 2017](#))) and exposes a series of workflows and utilities for geometry optimization, frequency analysis, one- and two-ended transition state search, and conformational search. When possible, these routines are implemented in a calculator-independent way, and almost all of these can be arbitrarily chained to one another to define a complex workflow.

The software was born out of necessity over the last five years of research in asymmetric catalysis: FIRECODE was designed to streamline the necessary use of multiple software and routines in the computational modeling of small molecules, in a single centralized hub. The program can be called via command line with a minimal plain text input file defining the workflow, or used as a Python library in Jupyter notebooks. The modular nature of the software enables rapid addition of new features, facilitating external contributions, while also retaining a small installation footprint for the core library. External programs are easily interfaced via API calls or file-based I/O. The concise input syntax also offers a streamlined definition of workflows and software orchestration with no coding experience required.

State of the field

On top of computational chemistry software, workflow utilities orchestrating sequential calculation steps are increasingly recognized as an essential asset in the streamlining of modeling efforts. Recently, version 6.0 of ORCA ([Neese et al., 2020; Neese, 2022; Tetenberg et al., 2026](#)) (free for academic use) started offering compound job functionality, which enables the chained execution of some core ORCA modules. The code is however not open source and functionality is limited to what is included in the ORCA program (*i.e.* no MLIPs, no similarity

39 pruning, no ensemble processing...). While ORCA workflows are more extensively implemented
40 in WEASEL ([Faccts](#)), this is commercial, closed-source software.

41 Related software to the one presented here include AQME, ([Alegre-Requena et al., 2023](#)) which
42 has what is perhaps the most similar design philosophy. Key differences with FIRECODE, besides
43 code architecture, lie in: 1) the main interaction platform of AQME happening via Jupyter
44 notebooks, while the software of this work favors the use of plain text input files, keeping
45 the inputs as minimal as possible. 2) The present software having a reactivity-centered set of
46 features, chiefly via fast semiempirical and ML calculators, over AQME's focus on DFT and on
47 the generation of ensemble properties like spectra and molecular descriptors.

48 Other workflow software emerging from the material chemistry community includes atomate2
49 ([A. Ganose et al., 2025](#); [A. M. Ganose et al., 2025](#)), AiIDA ([AiIDA — Computational Science
Workflow Manager — Aiida.net](#); [Huber et al., 2020](#); [Pizzi et al., 2016](#); [Uhrin et al., 2021](#)) and
50 QuAcc ([GitHub - Quantum-Accelerators/Quacc](#)).
51

52 Reactivity-centered software, primarily for the automated location of transition states, is also
53 gaining popularity. Examples include autodE ([GitHub - DuarteGroup/AutodE](#); [Young et al.,
2021](#)) and ML-FSM ([GitHub - TheGomeslab/ML-FSM](#); [Marks & Gomes, 2025](#)). The latter is
54 indeed interfaced as a utility in FIRECODE, given its light weight and flexibility in working
55 with different ASE calculators.
56

57 Other examples of related software, specifically targeting the exposure of high-level primitives
58 over structured workflows, include Cuby ([GitHub - Berquist/Cuby](#); [Řezáč, 2016](#)) (MD and
59 high-level DFT) written in Ruby, and the popular ASE ([Ase / Ase · GitLab — Gitlab.com](#);
60 [Bahn & Jacobsen, 2002](#); [Larsen et al., 2017](#)), which is extensively used as a lower level interface
61 for interacting with calculators in FIRECODE.

62 The development of this software from scratch was motivated by multiple reasons. First, a
63 modern and flexible hub for running computational chemistry workflows, particularly tailored
64 to modeling chemical reactivity and MLIPs, was lacking. In this context, we refer to “reactivity
65 workflows” as the sequence of sequential steps from a three-dimensional molecular representation
66 to a conformationally comprehensive characterization of bond breaking and forming events.
67 This requires the access to conformational search modules, efficient multistage ensemble
68 optimization routines, similarity pruning, the ability to use arbitrary geometrical constraints
69 throughout, the ability to optimize to local minima and saddle points alike, and the ability
70 to carry out vibrational analysis and calculate related thermochemical corrections to access
71 enthalpy/entropy/free energy values.

72 While the benefit of developing standalone applications for specific tasks remains, their
73 integration in a single hub is uniquely enabling for automation, and contributes to the
74 popularization of robust atomistic modeling. Second, modern machine-learned interatomic
75 potentials (MLIPs) are often centered around Python libraries like torch ([GitHub -
Pytorch/Pytorch](#)), benefitting greatly from a Python-based workflow manager. Additional
76 features of note of this software also include an implicit delta solvation scheme for gas
77 phase-trained MLIPs ¹, and numerical quasi-RRHO thermochemistry ([Grimme, 2012](#)).
78

79 Software design

80 The software is developed with the philosophy of maintaining the greatest amount of
81 functionality and module interoperability with the minimal list of dependencies, which should
82 ideally be as modern and lean as possible. All core dependencies are distributed via pypi,

¹That is, a calculator providing the energy and gradient differences between an atomic structure in an implicit solvent and the same structure in vacuum. These differential contributions are then added to the energy and gradient components of calculations that have no native implementation of solvation models (i.e. gas phase-trained MLIPs).

83 keeping the installation of the minimal version of the program down to a single command (uv
84 pip install firecode).

85 Non-essential dependencies can be installed based on the desired calculator and interfaces to
86 be used, all via conda.

87 The software implements various calculators (xTB (Bannwarth et al., 2019), tblite ([GitHub - Tblite/Tblite](#); Katbashev et al., 2025), AIMNET2 (Anstine et al., 2024), UMA (Wood et al., 2025)) via ASE ([Ase / Ase · GitLab — Gitlab.com](#); Bahn & Jacobsen, 2002; Larsen et al., 2017), which can be used interchangeably in various ensemble routines. Other core libraries leveraged throughout the codebase are numpy ([GitHub - Numpy/Numpy](#)) for algebraic manipulations, networkx ([GitHub - Networkx/Networkx](#); Hagberg, Schult, et al., 2008; Hagberg, Swart, et al., 2008) for graph utilities and prism_pruner ([GitHub - Ntampellini/Prism_pruner](#)) for the removal of duplicates. Sella ([GitHub - Zadorlab/Sella](#); Hermes et al., 2019, 2021, 2022) provides a saddle point optimizer, and rdkit ([GitHub - Rdkit/Rdkit](#); Greg Landrum, 2026) is used for ETKDG (Riniker & Landrum, 2015) conformational searches and SMARTS substructure matching. The ML-FSM library ([GitHub - Thegomeslab/ML-FSM](#); Marks & Gomes, 2025) provides a two-ended TS search utility complementing ASE's implementation of the NEB method (Ásgeirsson et al., 2021).

100 Other standalone conformational search software is instead interfaced via subprocess calls
101 and file-based I/O, like CREST (Pracht et al., 2020, 2024), ORCA's GOAT (Souza, 2025) and
102 racerts ([GitHub - Digital-Chemistry-Laboratory/Racerts](#); Schmid et al., 2026).

103 The modular nature of the calculators and interfaces allows the definition of complex workflows
104 in plain text files (input.txt):

```
firecode input.txt -n test_run
```

105 A second standalone executable exposes the core optimizer routines acting directly on structure
106 files:

```
firecode_opt conformers.xyz --opt --freq --solvent toluene [...]
```

107 Contributions by way of adding new routines (or interfaces) to the codebase is made
108 straightforward by design: individual "operators" (routines) are simply functions reading a
109 molecular .xyz file and returning the filename of the processed ensemble. Run information
110 can be accessed from the Embedder class or via environment variables.

```
def center_operator(filename: str, embedder: Embedder) -> str:
    """Example operator centering the molecule."""

    # the Embedder class stores global information
    embedder.avail_gpus # 1
    embedder.options.solvent # "ch2cl2"
    embedder.options.T # 298.15

    # get a copy of the molecule of interest
    mol = embedder.mols[filename]

    # center coordinates
    mol.coords[0] -= np.mean(mol.coords[0], axis=0)

    # save any data you might need later
    embedder.options.centered = True
    embedder.options.last_operator = "center"

    # write to global log
    embedder.log(f"--> Center operator: centered molecule {mol.basename}")
```

```
# write outfile and return its name
outfile = f"{mol.basename}_centered.xyz"
mol.to_xyz(outfile)

return outfile
```

Research impact statement

Since its earliest versions in 2021 (formerly [TSCoDe](#)), this software proved essential in orchestrating asymmetric catalysis workflows with ensembles up to thousands of structures ([Huth et al., 2024](#); [Marvich et al., 2026](#); [Rozema et al., 2025](#); [Tampellini et al., 2024, 2025](#); [Tampellini, Choi, et al., 2025](#)). These included the generation of large conformational ensembles for flexible catalysts and their non-covalent aggregates (often including specific constraints) and the multilevel optimization and pruning of these ensembles to a smaller number of non-redundant structures. These refined ensembles were then directed to higher-level DFT optimization to achieve quantitative insight into stereoselectivity (*i.e.* a predicted $\Delta\Delta G^\ddagger$ value) and obtain other conformationally-comprehensive properties (*i.e.* flexibility, relative populations, number of interaction topologies).

Adoption of FIRECODE is starting to grow, with pypi reporting ~250 downloads per month at the present time. Part of the core code was exported to the prism_pruner library ([GitHub - Ntampellini/Prism_pruner](#)) to facilitate its adoption by the computational chemistry startup Rowan ([Rowan | ML-Powered Molecular Design and Simulation — Rowansci.com](#)) (not affiliated with the author).

AI usage disclosure

Generative AI tools (Claude, Gemini) were used in some cases to generate first drafts of Python functions that were then manually curated line-by-line and interfaced with the rest of the codebase by hand. No generative AI tools were used in the writing of this manuscript.

Acknowledgements

We acknowledge years of research support from both Alma Mater Studiorum - University of Bologna (Prof. Giorgio Bencivenni and Prof. Paolo Righi) and Yale University (Prof. Scott J. Miller). These environments provided both the computational resources to run and develop the software as well as the primary problems it was designed to address and was stress-tested against.

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